DELAYED METROLOGY

Keep adapting.

REFLEX

Local calibration while measurements are in flight.

Double Zero

Double Zero

A label describes a past process state.

First deployment hypothesis

- One semiconductor equipment family, one quality target
- Known sample and measurement-arrival timestamps
- Shadow evaluation before production decisions

The model must distinguish

- Persistent drift from transient measurement noise
- Source time from arrival time
- A useful correction from an unreliable one

IMPLEMENTED LOCALLY

Double Zero

Learn only from measurements that arrived.

1

Retain time

Store the original sample timestamp and predictions.

2

Update states

Maintain GRAPE, Kalman and Bayesian regime models.

3

Weight evidence

Update model weights using arrived-label errors.

Fixed every-fifth sampling. 32 regime hypotheses max. No future-label oracle.

DATASET 1 · FIVE-SAMPLE DELAY

Double Zero

12.05% lower P90 error.

Metric	Delayed GRAPE	REFLEX ensemble	Difference
Requests / 542 samples	108 or 109	**Same schedule**	**Equal budget**
Mean RMSE	0.3257	**0.3142**	**3.53% lower**
Mean P90 absolute error	0.3958	**0.3481**	**12.05% lower**

Five offsets. P90 conditional 95% block interval: 1.78–18.80%. Exploratory, not a fresh blind test.

ALL DELAYED CONDITIONS · POSITIVE = IMPROVEMENT

Double Zero

The 10% RMSE target is still unmet.

Dataset / delay	RMSE change	P90 change	Assessment
1 / 5 samples	+3.53%	+12.05%	Strongest tail result
1 / 20 samples	−0.12%	+7.61%	RMSE regresses
2 / 5 samples	+0.52%	+0.55%	Small change
2 / 20 samples	+0.20%	+0.07%	Uncertain change

Mean RMSE improvement: 1.03%. Repeatedly inspected data; intervals do not correct selection bias.

MEASURED CPU EXECUTION

Double Zero

A compact model beside the equipment.

Step includes both candidate and baseline plus arrived-label updates; excludes sensors and instruments.

4.3 KiB

Dataset 1 artifact

33.6 KiB

Dataset 2 artifact

0.084 ms

M5 Pro Dataset 2 step p95

COMMERCIAL HYPOTHESES · NO VALIDATED PRICING

Double Zero

Sell a measured validation outcome.

Paid pilot

KRW 5–10 million
Fresh timestamped stream. Fixed measurement budget.

Integration

KRW 20–50 million
Local runtime. Traceable model updates.

Scale path

Equipment/OEM licensing
Only after repeatable field value.

Potential moat: permitted process-transition data and reusable adapters. Not established yet.

Double Zero

One fresh process. One honest proof.

Seeking a semiconductor process-data partner.

Validate real delays, large errors and engineering time.

SOURCES

GRAPE: arxiv.org/abs/2611.00806

Merging inspiration: arxiv.org/abs/2611.03230